1 Basic concepts

Whenever we introduce a new concept we will first write the rigorous definition, and right after give an intuitive interpretation.

1.1 Categories

Definition 1.1.1. *A* category $\mathbf{C}$ *consists of*

- *A collection* $\mathbf{C}_0$*, whose elements are called the* objects *of* $\mathbf{C}$ *and are usually denoted by uppercase letters* $X, Y, Z, \ldots$
- *A collection* $\mathbf{C}_1$*, whose elements are called the* morphisms *or* arrows *of* $\mathbf{C}$ *and are usually denoted by lowercase letters* $f, g, h, \ldots$

such that

- *Each morphism has assigned two objects, called* source *and* target*, or* domain *and* codomain*. We denote the source and target of the morphism* f *by* $s(f)$ *and* $t(f)$*, respectively. If the morphism* f *has source* X *and target* Y*, we also write* $f : X \to Y$*, or more graphically,* $X \xrightarrow{f} Y$*.*
- *Each object* X *has a distinguished morphism* $\mathrm{id}_X : X \to X$*, called* identity morphism*.*
- *For each pair of morphisms* f, g *such that* $t(f) = s(g)$ *there exists a specified morphism* $g \circ f$*, called the* composite morphism*, such that* $s(g \circ f) = s(f)$ *and* $t(g \circ f) = t(g)$*. More graphically:*

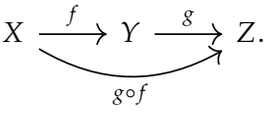
$$f : X \to Y\,,\ g : Y \to Z \qquad \rightsquigarrow \qquad g \circ f : X \to Z.$$

These structures need to satisfy the following axioms.

- *Unitality: for every morphism* $f : X \to Y$*, the compositions* $f \circ \mathrm{id}_X$ *and* $\mathrm{id}_Y \circ f$ *are both equal to* f*.*
- *Associativity: for* $f : X \to Y$*,* $g : Y \to Z$*, and* $h : Z \to W$*, the compositions* $h \circ (g \circ f)$ *and* $(h \circ g) \circ f$ *are equal.*

A category is a very general structure, the objects and morphisms can be anything, provided that they satisfy the properties given above. Below we will give some examples of categories arising from standard mathematical practice.

Caveat. The composite $g \circ f$ means *first* applying f, then g:

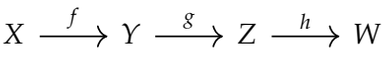


This choice of order, which is unfortunately incompatible with diagrams, comes from traditional mathematics: in expressions such as

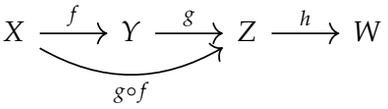
$$\sin(\cos(x))$$

one must first take the cosine, and then the sine. A way to avoid confusion is to read $g \circ f$ as "g *after* f".

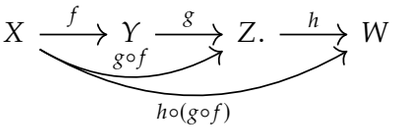
Here is a depiction of associativity: given three composable morphisms



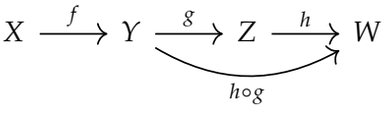
we have a priori two different ways to compose them. We could first compose f and g,



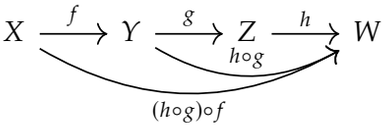
and then compose the resulting $g \circ f$ with h, to obtain



Alternatively, we could first compose g with h,



and then compose f with the resulting $h \circ g$, to obtain



The associativity condition is then saying that the two methods give the same result, that is, $h \circ (g \circ f) = (h \circ g) \circ f$. We can therefore write the triple composition without brackets, $h \circ g \circ f$, without any ambiguity.

A first intuitive way to think about a category is the following.

1.1.1 Categories as relations

Here is the first intuitive idea of what a category is.

Idea. *A category is a collection of objects which are related to each other in a consistent way.*

Let's see what this means by looking at examples.

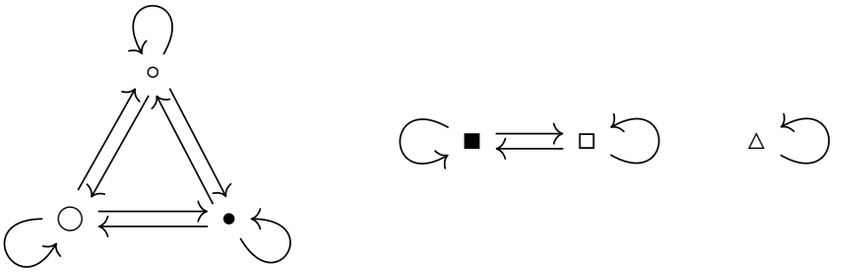
Example 1.1.2 (sets and relations)**.** An *equivalence relation* on a set X is a relation $\sim$ satisfying the following properties.

(a) Reflexivity: for each $x \in X$ we have $x \sim x$;

(b) Transitivity: for each $x, y, z \in X$, if $x \sim y$ and $y \sim z$, then $x \sim z$;

(c) Symmetry: for each $x, y \in X$, if $x \sim y$ then $y \sim x$ as well.

An equivalence relation is a mathematical way of formalizing the idea of "having something in common", such as "having the same shape". For example, here is what we obtain if, in a set of shapes, we draw an arrow whenever we want to say "has the same shape as" (regardless of color and size):



We note that:

(a) x has the same shape as itself;

(b) If x has the same shape as y and y has the same shape as z, then x has the same shape as z;

(c) If x has the same shape as y, then y has the same shape as x.

An equivalence relation defines a category in the following way. It intuitively looks like the picture above.

- The objects are the elements of X;
- Given $x, y \in X$, there exists a unique morphism $x \to y$ if and only if $x \sim y$.

The identity at an object x is the unique arrow $x \to x$ given by $x \sim x$. We know there is one, by reflexivity, and by definition there is only one, so we don't have to worry about *which* arrow $x \to x$ is the distinguished one. Just as well, the composition is given by transitivity: if we have arrows $x \to y$ and $y \to z$ it means that $x \sim y$ and $y \sim z$, and so by transitivity $x \sim z$ as well, which means that there is an arrow $x \to z$. Again, we don't have to worry about which arrow $x \to z$ we want, since there is only one. The symmetry property says that if there is an arrow $x \to y$, then there is an arrow $y \to x$ as well. In other words, "we can always go back". This last property is not important to have a category, it is merely an extra property that this category will have.

Example 1.1.3 (sets and relations)**.** An *order relation* on a set X is a relation $\leq$ satisfying the following axioms.

(a) Reflexivity: for each $x \in X$ we have $x \leq x$;

(b) Transitivity: for each $x, y, z \in X$, if $x \leq y$ and $y \leq z$, then $x \leq z$;

(c) Antisymmetry: for each $x, y \in X$, if $x \leq y$ and $y \leq x$, then necessarily $x = y$.

An order relation is a mathematical way of formalizing the idea of "comparing sizes", such as "being smaller or equal". For example:

(a) x is smaller or equal to itself;

(b) If x is smaller or equal to y and y is smaller or equal to z, then x is smaller or equal to z;

(c) If x is smaller or equal to y and y is smaller or equal to x, then $x = y$.

An order relation defines a category in an analogous way to what happens with equivalence relations: objects are elements of X, and there is a unique arrow $x \to y$ if and only if $x \leq y$. Again, reflexivity and transitivity are enough to have a category structure, antisymmetry is an additional property.

The most general relation which gives rise to a category is called a preorder, and it has no conditions on symmetry.

Example 1.1.4 (sets and relations, economics)**.** A *preorder relation* on a set X is a relation $\lesssim$ satisfying only the axioms:

(a) Reflexivity: for each $x \in X$ we have $x \lesssim x$;

(b) Transitivity: for each $x, y, z \in X$, if $x \lesssim y$ and $y \lesssim z$, then $x \lesssim z$.

Again, a preorder defines a category where there is a unique morphism between x and y if and only if $x \lesssim y$, with identities and composition given by reflexivity and transitivity. Equivalence relations and order relations are both special cases of preorder relations. A non-trivial example of a preorder is the *price*, in economics, i.e. the relation of being "cheaper or equally priced". For example:

(a) x has lower or equal price to itself;

(b) If x has lower or equal price to y and y has lower or equal price to z, then x has lower or equal price to z.

However, differently from a partial order, if x has lower or equal price to y and y has lower or equal price to x we don't have that necessarily $x = y$: they will simply have the same price. They are not the same object, but they are equivalent: you can trade one for the other one and vice versa.

Exercise 1.1.5 (sets and relations)**.** Let $(X, \lesssim)$ be a preorder. Prove that the relation $\sim$ given by

$$x \sim y \text{ if and only if } x \lesssim y \text{ and } y \lesssim x$$

is an equivalence relation.

For all these examples, given two objects, there is at most one morphism between them. This is not the case in general, and sometimes the *choice* of the morphism matters.

1.1.2 Categories as operations

Here is another way to think about categories.

Idea. *A category is a collection of operations which can be composed in a consistent way.*

Here are some examples of categories where this interpretation is most helpful. We encourage the readers unfamiliar with group theory *not* to skip the examples below: once one gets the basic idea of a group, many concepts in category theory are much easier to understand.

Example 1.1.6 (group theory)**.** A *group* is a nonempty set G together with

(a) A distinguished element $1 \in G$ called the *neutral element* or *unit*;

(b) A binary operation $G \times G \to G$ called *multiplication*, which we denote by $(g, h) \mapsto g \cdot h$;

(c) For each element $g \in G$ an element g^{-1} called the *inverse*;

such that the following properties hold.

(a) Unitality: for each $g \in G$, the multiplications $g \cdot 1$ and $1 \cdot g$ are both equal to g;

(b) Associativity: for each $g, h, i \in G$, the multiplications $(g \cdot h) \cdot i$ and $g \cdot (h \cdot i)$ are equal;

(c) Inverse law: for each $g \in G$, the multiplications $g \cdot g^{-1}$ and $g^{-1} \cdot g$ are both equal to 1.

A group in mathematics is used to model the *symmetries* of some structures, namely the ways in which we can act on some object while keeping it "the same". For example, rotations of the plane form a group, and also permutations of a set. A group is similar to a category: there is the neutral element 1, which behaves like an identity, and there is the multiplication, which behaves like composition (it is even associative). How can we turn then a group into a category? An important feature of a group G is that *we can compose any two elements of* G. There is no requirement of "matching source and target", as there is for a category. Therefore, in order to view a group as a particular category, we should make sure that the source and target of all morphisms always match. The only way to do this in general is to have *a unique object*. Here is the construction in detail.

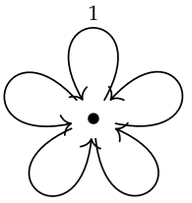
Definition 1.1.7. *Let G be a group. The* delooping *of G, denoted by $\mathbf{B}G$, is the following category.*

- *There is a single object $\bullet$;*

- *There is a morphism $\bullet \to \bullet$ for each element $g \in G$. We denote the morphism with the same letter, such as $g : \bullet \to \bullet$.*
- *The identity of the object $\bullet$ is the morphism given by $1 \in G$;*
- *The composition is given by the multiplication of G. That is, the composition $g \circ h$ of the morphisms given by $g, h \in G$ is the morphism given by the element $g \cdot h$ of G.*

Since associativity and unitality hold for elements of G, they also hold for the morphisms of $\mathbf{B}G$. Therefore, $\mathbf{B}G$ is a category.

Graphically, $\mathbf{B}G$ looks like a point with loops, with one special loop (the identity):



The term "delooping" comes from algebraic topology. There is a deep connection between algebraic topology and category theory, which is widely explored on the nLab, and the delooping is part of this connection. For us, however, it will be merely a way of obtaining a category from a group (or monoid, see below).

Remark 1.1.8. The category $\mathbf{B}G$ contains the same information as the group G, it is just a different way to express it. This is why some authors prefer omitting the symbol $\mathbf{B}$ and simply say that "a group is a certain category with a single object". In this course we will keep the two notions distinct. Remember, however, that both encode the same data.

Remark 1.1.9. Since a group is usually modeling the "symmetries of something", sometimes we can make this delooping construction more concrete. For example, for the group of rotations of $\mathbb{R}^2$, we can take the object $\bullet$ to be $\mathbb{R}^2$ and the arrows to be rotations $\mathbb{R}^2 \to \mathbb{R}^2$ (as linear maps). This will be made precise in the future. However, for now, the construction is purely formal.

In order to form the category $\mathbf{B}G$ from the group G we never used the inverse elements and the inverse law. These, just as the symmetry property of equivalence relations, are not needed to have a category, and merely give to $\mathbf{B}G$ an additional structure, namely an "inverse" morphism g^{-1} to each morphism g (more on this in 1.1.5). Therefore, if we drop the inverse requirement, we can still get a category.

Example 1.1.10. A *monoid* is a nonempty set M together with

(a) A distinguished element $1 \in M$ called the *neutral element* or *unit*;

(b) A binary operation $M \times M \to M$ called *multiplication*, which we denote by $(g, h) \mapsto g \cdot h$;

such that the following properties hold.

(a) Unitality: for each $g \in M$, the multiplications $g \cdot 1$ and $1 \cdot g$ are both equal to g;

(b) Associativity: for each $g, h, i \in M$, the multiplications $(g \cdot h) \cdot i$ and $g \cdot (h \cdot i)$ are equal.

The *delooping* of a monoid M is the category $\mathbf{B}M$ with, just as for groups, a single object $\bullet$, and as morphisms the elements of M with their identity and composition.

Just as a group is used to model the symmetries of some object, a monoid is used to model the transformations of an object which are not necessarily invertible. For example consider, instead of only the rotations of the plane $\mathbb{R}^2$, the set of *all linear maps* $\mathbb{R}^2 \to \mathbb{R}^2$, including the one that maps everything to $(0, 0)$. These maps include the identity and can be composed, so they form a monoid. However, not all of them are invertible, so they don't form a group. There are many examples similar to this:

- Given a vector space V, all linear maps $V \to V$ form a monoid;
- Given a topological space X, all continuous maps $X \to X$ form a monoid;
- Given a set X, all functions $X \to X$ form a monoid;
- ...and so on.

Here is another couple of examples from probability theory:

- Given a measurable space X and a Markov kernel $k : X \to X$, the set given by the identity id_X and the repeated applications of k, that is, the set
$$\{\mathrm{id}_X,\ k,\ k^2,\ k^3,\ \dots\}$$
is a monoid. Sometimes this is called the *Markov semigroup* generated by k.
- More generally, given a measurable space X, all Markov kernels $X \to X$ form a monoid. Sometimes this is called the *full Markov semigroup on* X.

Remark 1.1.11. In fields such as probability theory, people sometimes prefer the notion of "semigroup" to the one of monoid. A semigroup is like a monoid, but without the unit (and the unitality property). In category theory, by convention, we always use monoids. The difference is minimal, one can almost always add the unit, since it corresponds to "doing nothing", and it is largely only a matter of convention. (Requiring the identity is analogous to admitting the zero in the set of natural numbers.)

We conclude this part with an exercise.

Exercise 1.1.12 (important!)**.** Prove that, in a group or in a monoid, the neutral element is unique. In other words, if both 1 and 1′ satisfy the properties of the neutral element, then $1 = 1'$.

1.1.3 Categories as spaces and maps with extra structure

In the case of the categories defined by equivalence relations, orders, and preorders, we didn't need to worry about the equality of composite arrows (for example to check associativity), since between any two objects, there is at most one arrow. In categories arising as delooping of groups or monoids, we didn't need to worry about matching source and target of arrows, since there is only a single object. In the most general case, one needs always to check both. Here is another idea, which is helpful to study the categories of this more general kind.

Idea. *A category is a collection of sets or spaces equipped with extra structure, and maps between them which are compatible with that structure.*

Categories of this form are usually named after their objects.

Example 1.1.13 (sets and relations)**.** The category **Set** is the category whose objects are sets, and whose morphisms are maps (functions) between them.

For each set X, the identity function is a function $X \to X$. Functions can be composed, and the composition is associative. Therefore **Set** is a category.

Example 1.1.14 (several fields)**.** Analogously to the category **Set**, the following are categories.

- **Top** has topological spaces as objects and continuous maps as morphisms.
- **Mfd** has smooth manifolds as objects and smooth maps as morphisms.
- **Vect** has vector spaces as objects and linear maps as morphisms.
- **Meas** has as objects measurable spaces, and as morphisms measurable maps.
- **Grp** has groups as objects and group homomorphisms as morphisms, i.e. maps $f : G \to H$ such that for each $g, g' \in G$, we have that $f(g \cdot g') = f(g) \cdot f(g')$.
- **Poset** has partially ordered sets as objects, and monotone maps as morphisms, i.e. maps $f : X \to Y$ such that for each $x, x' \in X$ with $x \leq x'$ in X, we have that $f(x) \leq f(x')$ in Y.

A category with sets as objects may have many choices of morphisms, for example, instead of functions, we may choose injective functions, or bijections, or even relations. The choice of morphisms reflects the choice of *context* which we want to consider, or the choice of *structure* that we want to preserve (and study).

Example 1.1.15 (graph theory)**.** For graphs there are many choices of morphisms, depending on what one wants to do with graphs (graph theory is so general and versatile that many choices are meaningful). For example, for undirected, unweighted graphs, a possible choice of morphisms $f : G \to H$ is functions between the sets of vertices which preserve the adjacency relation: if the vertices x and y are connected by an edge in G, then $f(x)$ and $f(y)$ must be connected by an edge in H. With this choice of morphisms, graphs and these morphisms form a category. This is however only one of many possible choices.

Exercise 1.1.16. In your field of mathematics (or physics, chemistry, computer science, economics,...) which structures do you work with the most? Can you construct a category with such structures? Are those structures better represented as objects or as morphisms of a category?

Here is also an example of what is *not* a category, because composition fails.

Example 1.1.17 (calculus)**.** Consider the following convex functions on $\mathbb{R}$:

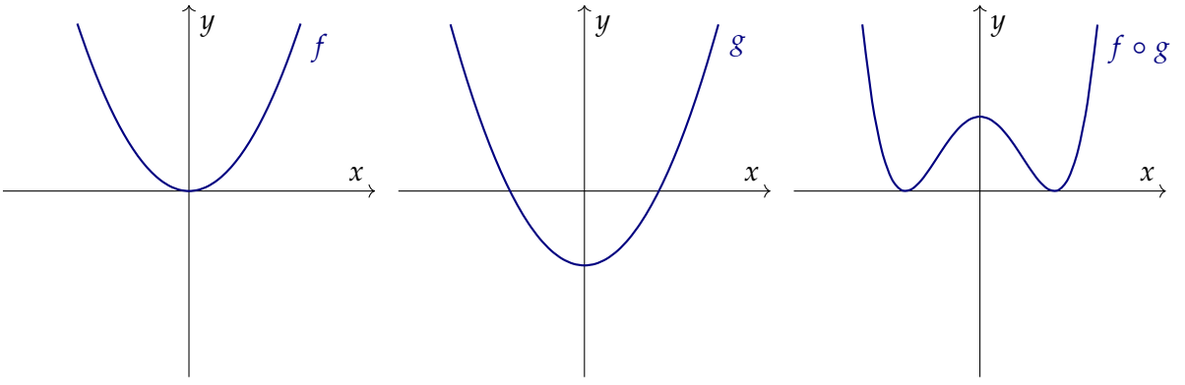
- $x \mapsto f(x) \coloneqq x^2$;

- $x \mapsto g(x) \coloneqq x^2 - 1$.

Both are even strictly convex (their graphs are parabolas, see the picture below). However, the composite $f \circ g : \mathbb{R} \to \mathbb{R}$ is not convex: we have

$$f(g(x)) = (x^2 - 1)^2,$$

which is equal to 0 for $x = \pm 1$, and equal to 1 for $x = 0$.



Therefore, convex functions on $\mathbb{R}$ do not form a category. The same is true for lower semicontinuous maps.

Exercise 1.1.18 (analysis)**.** Show that there exists two lower semicontinuous maps $f, g : \mathbb{R} \to \mathbb{R}$ such that $f \circ g$ is not lower semicontinuous.

Therefore, whenever you define a category, check that all the axioms hold. Not everything is automatically a category.

Exercise 1.1.19 (analysis)**.** Can we give an additional requirement to convex maps, in such a way to make them closed under composition? Will they form a category that way? What about lower semicontinuous maps?

1.1.4 Set-theoretical considerations

The content of this part is a bit technical, it will not be really needed to understand the rest of the course. It should nevertheless be mentioned, in order to have a rigorous treatment and to avoid possible confusion. Moreover, the notation given in Definition 1.1.23 will be used in the future.

You may know from your set theory knowledge that there is no such thing as the "set of all sets" (the reason being: does that set contain itself?). Therefore, if we want a "category of all sets", the objects of this category cannot form a set. That's why, in the definition of a category (Definition 1.1.1) we said *collections* rather than sets: the objects and morphisms may in general be proper classes.

Definition 1.1.20. *A category* $\mathbf{C}$ *is called* small *if* $\mathbf{C}_0$ *and* $\mathbf{C}_1$ *are sets.*

A category $\mathbf{C}$ *is called* locally small *if for every two objects* X *and* Y *of* $\mathbf{C}$*, the morphisms* $X \to Y$ *form a set.*

Alternatively, one may speak of universes and *small and large sets* instead of sets and proper classes. In this formalism, a category is small if $\mathbf{C}_0$ and $\mathbf{C}_1$ are small sets, and locally small categories are defined analogously.

Most categories of interest are locally small. In particular, all the examples of categories analyzed so far are locally small. By construction, moreover, the categories obtained from relations on a set or as delooping of a monoid are small. You can prove converses of this last statement by exercise:

Exercise 1.1.21 (sets and relations)**.** Prove that a small category with at most a single arrow between any two objects is a preorder. (Hint: there is not much to prove.)

Just as well:

Exercise 1.1.22. Prove that a locally small category with a single object is a monoid.

We conclude this section with a useful piece of notation, which we will use in the rest of this course.

Definition 1.1.23. *Let X and Y be objects of a locally small category $\mathbf{C}$. The* hom-set *or* hom-space *of X and Y is the set of morphisms of $\mathbf{C}$ from X to Y. We will denote it by* $\mathrm{Hom}_{\mathbf{C}}(X, Y)$.

From now on we will mostly work with locally small categories.

Caveat. Sometimes a mathematical structure can be an object of different categories. For example $\mathbb{R}$ and $\mathbb{R}^2$ are sets, topological spaces, vector spaces, and so on. This is why it is important to keep track of the category when we write the hom-sets. For example,

- $\mathrm{Hom}_{\mathbf{Set}}(\mathbb{R}, \mathbb{R}^2)$ is the set of *all* functions $\mathbb{R} \to \mathbb{R}^2$, i.e. the morphisms of **Set**;
- $\mathrm{Hom}_{\mathbf{Top}}(\mathbb{R}, \mathbb{R}^2)$ is the set of all *continuous* functions $\mathbb{R} \to \mathbb{R}^2$, i.e. the morphisms of **Top**;
- $\mathrm{Hom}_{\mathbf{Vect}}(\mathbb{R}, \mathbb{R}^2)$ is the set of all *linear* functions $\mathbb{R} \to \mathbb{R}^2$;
- . . . and so on.

If you want to know more about the set-theoretical issues in category theory, you can read these notes by Mike Shulman, [Shu08]. However I recommend you read them after learning a bit more about category theory.

1.1.5 Isomorphisms and groupoids

For the case of equivalence relations, and for delooping of groups, we had categories with a special property, which intuitively said that "we can always go back". Let's now try to make this intuition precise.

Definition 1.1.24. *Let X and Y be objects in a category $\mathbf{C}$. An* isomorphism *is a pair of morphisms*

$$X \underset{g}{\overset{f}{\rightleftarrows}} Y$$

such that

(a) $g \circ f = \mathrm{id}_X$, *and*

(b) $f \circ g = \mathrm{id}_Y$.

If there exists an isomorphism between X *and* Y, *we say that* X *and* Y *are* isomorphic.

We see in this definition that g *not only* is in the opposite direction as f, but also "undoes f", in the sense that applying g after f is like doing nothing at all. And just as well, f undoes g.

Caveat. The two conditions do not imply one another, as the following example shows. Therefore, in order to have an isomorphism, both conditions need to be satisfied separately.

Example 1.1.25 (calculus, linear algebra)**.** Let $X = \mathbb{R}^2$ and $Y = \mathbb{R}$. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the projection onto the x-axis $(x, y) \mapsto x$, and let $g : \mathbb{R} \to \mathbb{R}^2$ be the inclusion of the x-axis $x \mapsto (x, 0)$. Then $f \circ g = \mathrm{id}_{\mathbb{R}}$, but $g \circ f \neq \mathrm{id}_{\mathbb{R}^2}$: in particular, $g \circ f(x, y) = g(x) = (x, 0)$, which is in general different from (x, y).

Exercise 1.1.26 (important!)**.** Let $f : X \to Y$ be a morphism in a category. Suppose that both g and $g' : Y \to X$ are inverse to f. Prove that $g = g'$. Conclude that, in particular, inverses in a group are unique.

Here are some examples of isomorphisms.

Example 1.1.27 (several fields)**.**

- In the category **Set**, the isomorphisms are the bijective functions.
- In **Top**, the isomorphisms are the homeomorphisms: invertible continuous maps whose inverse is continuous. Mind: this is strictly stronger than saying continuous and bijective (can you give a counterexample?).
- In **Mfd**, the isomorphisms are the diffeomorphisms.
- In **Vect**, the morphisms are bijective linear maps.

Exercise 1.1.28 (important!)**.** Prove that the composition of invertible morphisms is invertible.

Let's now generalize equivalence relations and groups, i.e. categories where "we can always go back".

Definition 1.1.29. *A* groupoid *is a category where all the morphisms are invertible.*

Here are some examples of groupoids.

Example 1.1.30 (sets and relations)**.** We have already seen equivalence relations. For an equivalence relation $\sim$ on a set X, we have that if $x \sim y$, i.e. we have an arrow $x \to y$, then $y \sim x$, i.e. we have an arrow $y \to x$. Checking that the two arrows invert each other is trivial, guaranteed by uniqueness. In detail, the composition $x \to y \to x$ has to be equal to the identity $x \to x$, because there is at most one arrow $x \to x$.

Example 1.1.31 (group theory)**.** The delooping $\mathbf{B}G$ of a group G is a groupoid (this is why it's called "groupoid": it generalizes a group). The inverses are given by the inverses in G.

Another example of a groupoid is the groupoid of sets and *bijections*, since those are exactly the invertible morphisms. We can do this more generally.

Definition 1.1.32. *Let* **C** *be a category. The* core *of* **C** *is the groupoid whose objects are the objects of C and whose morphisms are the isomorphisms of C.*

We have seen another example of a core already: in Exercise 1.1.5, we were canonically obtaining an equivalence relation from a preorder. Therefore, the core of a preorder is an equivalence relation.

We conclude this section with a "philosophical" remark. In category theory, one usually never talks about two objects being *equal*, but only *isomorphic*. In some sense, this reflects standard mathematical practice. For example, as vector spaces, every three-dimensional space is isomorphic to $\mathbb{R}^3$. But are they really *equal*? What would equality even mean in that context? What can be done in $\mathbb{R}^3$ can be done in any three-dimensional vector space, and that's what matters in the end. Therefore, isomorphism is the important notion between objects. When it comes to *morphisms*, on the other hand, two morphisms are often required to be *equal*. For example, in the definition of an isomorphism, $f^{-1} \circ f$ is *equal* to the identity id_X, not "isomorphic". Again, this reflects standard mathematical practice: functions between spaces can be equal. For example, two functions $f, g : X \to Y$ are equal if and only if for every $x \in X$ we have $f(x) = g(x)$.

One may be tempted to ask whether it is possible to also talk about *isomorphisms of maps* instead of equalities, so that objects and morphisms are treated more on an equal footing. This is exactly the subject of *higher category theory*, in which one can talk about morphisms between morphisms, and so on. Higher category theory is very helpful, for example, in algebraic geometry and algebraic topology. The nLab is the standard online wiki on this subject. In this course we will for the most part do *ordinary* category theory, where morphisms are allowed to be equal.

1.1.6 Diagrams, informally

One of the most powerful methods of category theory is *reasoning in terms of diagrams*. We start with an informal (but consistent) definition of a diagram. A more formal definition will be given later on in terms of functors.[1]

Definition 1.1.33 (informal)**.** *A* diagram *in a category* **C** *is a directed (multi-)graph formed out of objects and arrows of* **C** *such that:*

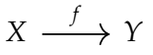
- *Each object and morphism may appear more than once in the diagram;*
- *Between any two objects there may be also more than one morphism;*
- *For each object X in the diagram, the identity is implicitly present in the diagram (but generally not drawn);*
- *For each composable edges (arrows) $f : X \to Y$ and $g : Y \to Z$ which are appearing head-to-tail in the diagram, the composite $g \circ f : X \to Z$ is implicitly present in the diagram (but generally not drawn).*

[1] See Section 1.4.5.

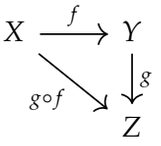
Since objects and morphisms may appear more than once in the diagram, as different vertices and edges, we will refer to vertices and edges to avoid ambiguity.

Exercise 1.1.34. Which morphisms are contained in the following diagrams, implicitly or explicitly? (Without assuming any additional equalities other than those required by our axioms.)

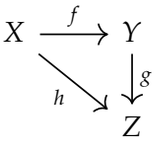
(a)



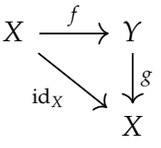
(b)



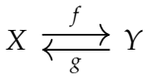
(c)



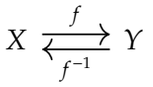
(d)



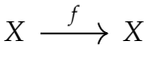
(e)



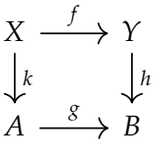
(f)



(g)



(h)

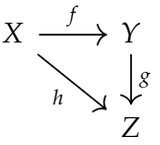


Definition 1.1.35. *A diagram is* commutative *(or it* commutes*) if for each two vertices X and Y in the diagram, all compositions along paths of composable arrows connecting X to Y are equal.*

Caveat. Not every diagram commutes!

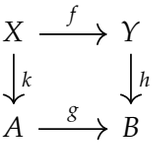
Example 1.1.36.

- The diagram



  is commutative if and only if $g \circ f = h$.

- The diagram



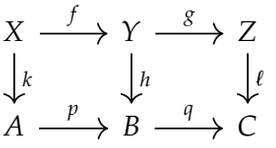
  is commutative if and only if $h \circ f = g \circ k$.

Exercise 1.1.37. Which diagrams in Exercise 1.1.34 commute? For those which do not commute, which morphisms should we require to be equal in order for the diagrams to be commutative?

Exercise 1.1.38 (sets and relations)**.** Prove that in a preorder (or in a poset, or in an equivalence relation) every diagram commutes.

Commutative diagrams can be pasted, as the following exercise shows.

Exercise 1.1.39. Consider the following diagram, made out of two adjacent squares:

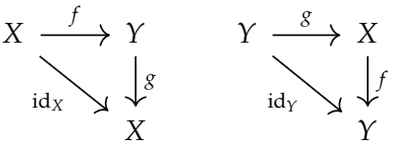


Suppose that the left square and the right square, separately, commute. Prove that the whole diagram then commutes.

More generally, commutative diagrams of any shape can be pasted to give again commutative diagrams.

Recall the definition of an isomorphism (Definition 1.1.24). We can rewrite it purely in terms of commutative diagrams.

Definition 1.1.40 (alternative)**.** *Two morphisms $f : X \to Y$ and $g : Y \to X$ of a category* **C** *are inverse to each other, forming an isomorphism, if and only if the following two diagrams are commutative.*



Again, keep in mind that both diagrams need to be checked separately, the two conditions are independent.

1.1.7 The opposite category

Another useful technique in category theory is "reversing all the arrows".

Definition 1.1.41. *Let C be a category. The* opposite category *of* $\mathbf{C}$*, denoted as* $\mathbf{C}^{\mathrm{op}}$*, is the category obtained as follows.*

- *Objects are just the objects of* $\mathbf{C}$*;*
- *A morphism, denoted as* f^{op}*, between two objects X and Y is a morphism f in* $\mathbf{C}$ *from Y to X (mind the direction). Graphically:*

$$\begin{array}{c} X \\ \downarrow{\scriptstyle f^{\mathrm{op}}} \\ Y \end{array} \quad \rightsquigarrow \quad \begin{array}{c} X \\ \uparrow{\scriptstyle f} \\ Y \end{array}$$

- *The identity at each object X is given by* $\mathrm{id}_X{}^{\mathrm{op}}$*;*
- *Composition is the same as in* $\mathbf{C}$*, but the order is reversed:* $g^{\mathrm{op}} \circ f^{\mathrm{op}}$ *is defined to be* $(f \circ g)^{\mathrm{op}}$*. In diagrams:*

$$\begin{array}{c} X \\ \downarrow{\scriptstyle f^{\mathrm{op}}} \\ Y \\ \downarrow{\scriptstyle g^{\mathrm{op}}} \\ Z \end{array} \quad \rightsquigarrow \quad \begin{array}{c} X \\ \uparrow{\scriptstyle f} \\ Y \\ \uparrow{\scriptstyle g} \\ Z \end{array}$$

Here are examples on how the opposite category looks in practice.

Example 1.1.42 (sets and relations)**.** Let $(X, \leq)$ be a poset, i.e. a set equipped with a partial order. As we have seen in Example 1.1.3, this is in particular a category, with an arrow $x \to y$ if and only if $x \leq y$. The opposite category has then an arrow $x \to y$ if and only if $y \leq x$. We can consider it as the category obtained by the opposite relation, $\geq$ instead of $\leq$. For example, for $X = \mathbb{R}$ with its usual order, $(\mathbb{R}, \leq)^{\mathrm{op}} \cong (\mathbb{R}, \geq)$, ordered downward.

Example 1.1.43 (group theory)**.** Let G be a group. We have seen in Section 1.1.2 that the delooping $\mathbf{B}G$ is a category. The opposite category $(\mathbf{B}G)^{\mathrm{op}}$ will again have a single object, and morphisms will be again elements of G, however the composition is reversed: $(g \circ h)^{\mathrm{op}} = h^{\mathrm{op}} \circ g^{\mathrm{op}}$.

Since the axioms of categories are symmetric with respect to reversing all arrows, the following is true: *a statement in a category* $\mathbf{C}$ *is true if and only if the dual statement is true in* $\mathbf{C}^{\mathrm{op}}$.

Caveat. A statement is true in $\mathbf{C}$ if and only if the dual statement is true in $\mathbf{C}^{\mathrm{op}}$, but in general, the two statements do not look the same.

This is why this principle is so powerful: for each result we prove in category theory there is always a dual result that we get for free. Here is an example.

Exercise 1.1.44. Let $f, g : X \to Y$ be morphisms in $\mathbf{C}$. Prove that $f = g$ if and only if $f^{\mathrm{op}} = g^{\mathrm{op}}$ in $\mathbf{C}^{\mathrm{op}}$. (Hint: there is not much to prove.)

Given the result of the exercise, the following statements then just follow as corollaries.

Corollary 1.1.45. *A diagram in* $\mathbf{C}$ *commutes if and only if the dual diagram in* $\mathbf{C}^{\mathrm{op}}$ *commutes.*

As an example, the following diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow{\scriptstyle k} & & \downarrow{\scriptstyle h} \\ A & \xrightarrow{g} & B \end{array}$$

commutes if and only if $h \circ f = g \circ k$. This happens if and only if $f^{\mathrm{op}} \circ h^{\mathrm{op}} = k^{\mathrm{op}} \circ g^{\mathrm{op}}$, i.e. if the dual diagram commutes:

$$\begin{array}{ccc} X & \xleftarrow{f^{\mathrm{op}}} & Y \\ \uparrow{\scriptstyle k^{\mathrm{op}}} & & \uparrow{\scriptstyle h^{\mathrm{op}}} \\ A & \xleftarrow{g^{\mathrm{op}}} & B \end{array}$$

Moreover, since isomorphisms can be defined purely in terms of commutative diagrams (Definition 1.1.40), we get:

Corollary 1.1.46. *A morphisms* $f : X \to Y$ *is invertible if and only if* $f^{\mathrm{op}} : Y \to X$ *is invertible. (What will the inverse be?)*

In particular, if X *and* Y *are isomorphic in* $\mathbf{C}$*, then they are also isomorphic in* $\mathbf{C}^{\mathrm{op}}$*.*

In the next section we will see more examples of how reversing the arrows can give meaningful statements.

1.2 Mono and epi

So far, we have expressed in terms of diagrams the notion of isomorphism, which gives the idea of when two objects are, in some sense, "the same". This generalizes the notion of "bijective map" between sets, and gives the right notion of "sameness" in other categories (such as homeomorphisms between topological spaces). We now would like to present a diagrammatic approach to talk about subspaces and quotients, generalizing injective and surjective maps. In category theory these maps will be called *monomorphisms* (from the Greek work meaning "one", as in "one-to-one"), and *epimorphisms* (from the Greek word meaning "onto").

1.2.1 Monomorphisms

Definition 1.2.1. *A morphism* $m : X \to Y$ *in a category* $\mathbf{C}$ *is called a* monomorphism*, or* mono*, if the following holds. For every object* A *of* $\mathbf{C}$ *and for every pair of maps* $f, g : A \to X$*, i.e. fitting into the diagram*

$$A \underset{g}{\overset{f}{\rightrightarrows}} X \xrightarrow{m} Y$$

and such that $m \circ f = m \circ g$*, we have that* $f = g$ *(i.e. the whole diagram commutes).*